

OS LAB

PRACTICAL

1) Anonymous Pipes:

```
#include <stdio.h>

#include <unistd.h>

#include <string.h>

#include <sys/types.h>

int main()

{

    int fd[2];

    pid_t pid;

    char message[] = "Hello from Parent!";

    char buffer[100];

    // Create pipe

    pipe(fd);

    // Create child process

    pid = fork();

    if (pid > 0)

    {

        // Parent process

        close(fd[0]); // Close reading end

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]); // Close writing end
```

```

    }

    else if (pid == 0)
    {
        // Child process

        close(fd[1]); // Close writing end

        read(fd[0], buffer, sizeof(buffer));

        printf("Child received: %s\n", buffer);

        close(fd[0]); // Close reading end
    }

    return 0;
}

```

OUTPUT:

```

lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ ls -la
total 8
drwxr-xr-x 2 lahari lahari 4096 Aug 29 08:16 .
drwxr-xr-x 7 lahari lahari 4096 Aug 29 08:15 ..
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ cd ..
lahari@DESKTOP-AIC33F7:~/OSLab$ rmdir pipes.c
lahari@DESKTOP-AIC33F7:~/OSLab$ cd pipes
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ nano anonymouspipes.c
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ ls
anonymouspipes.c
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ gcc anonymouspipes.c -o anonymouspipes
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ ls
anonymouspipes  anonymouspipes.c
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ ./anonymouspipes
Child received: Hello from Parent!
lahari@DESKTOP-AIC33F7:~/OSLab/pipes$ |

```

2) Named Pipes / FIFO:

a) Create a FIFO

```
#include <stdio.h>
```

```
#include <sys/types.h>
```

```
#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");
    return 0;
}
```

b) FIFO_READ

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);

    read(fd, buffer, sizeof(buffer));

    printf("Received: %s\n", buffer);
}
```

```
close(fd);

return 0;
}
```

c. FIFO_write.c

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);

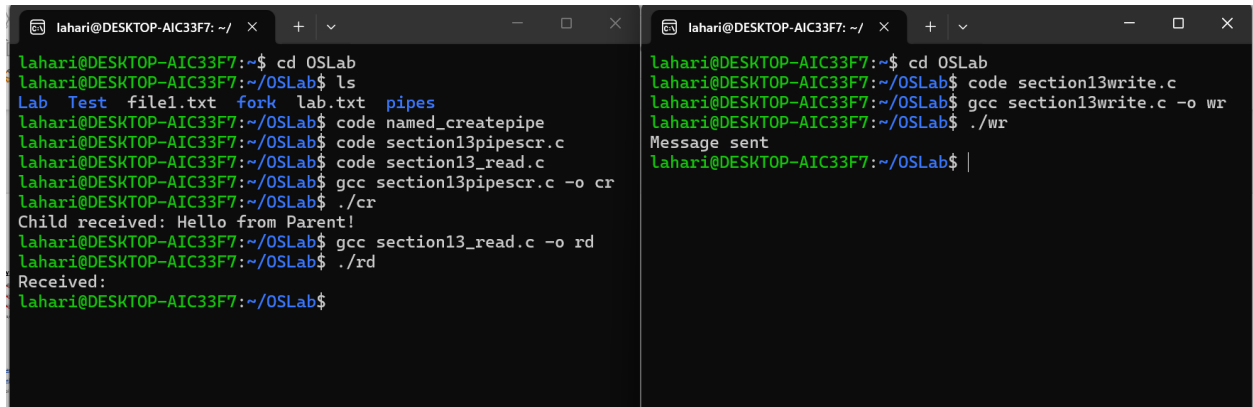
    write(fd, message, strlen(message) + 1);

    close(fd);

    printf("Message sent\n");
}
```

```
return 0;

}
```



The image shows two terminal windows side-by-side, both with the title bar 'lahari@DESKTOP-AIC33F7: ~/'. The left window shows a series of commands to create and compile a program using pipes. The right window shows the compilation and execution of a program that sends a message through a pipe.

```
lahari@DESKTOP-AIC33F7:~$ cd OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ ls
Lab Test file1.txt fork lab.txt pipes
lahari@DESKTOP-AIC33F7:~/OSLab$ code named_createpipe
lahari@DESKTOP-AIC33F7:~/OSLab$ code section13pipescr.c
lahari@DESKTOP-AIC33F7:~/OSLab$ code section13_read.c
lahari@DESKTOP-AIC33F7:~/OSLab$ gcc section13pipescr.c -o cr
lahari@DESKTOP-AIC33F7:~/OSLab$ ./cr
Child received: Hello from Parent!
lahari@DESKTOP-AIC33F7:~/OSLab$ gcc section13_read.c -o rd
lahari@DESKTOP-AIC33F7:~/OSLab$ ./rd
Received:
lahari@DESKTOP-AIC33F7:~/OSLab$
```

```
lahari@DESKTOP-AIC33F7:~$ cd OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ code section13write.c
lahari@DESKTOP-AIC33F7:~/OSLab$ gcc section13write.c -o wr
lahari@DESKTOP-AIC33F7:~/OSLab$ ./wr
Message sent
lahari@DESKTOP-AIC33F7:~/OSLab$
```